Chemistry Topic 8 — Nuclear Chemistry, Explosives and Environment

Covers syllabus (click to expand)

Radioactivity & Radiations (Alpha, Beta, Gamma Properties & Soddy-Fajans Displacement) | Half-Life Kinetics & Radiometric Dating (Carbon-14, Uranium-Lead, K-Ar) | Nuclear Fission vs Nuclear Fusion Thermodynamics | Hydrogen Bomb Principle | Master Chemical Explosives (Dynamite, TNT, RDX / C-4, PETN, ANFO, Gunpowder) | Acid Rain Chemistry & Marble Stone Leprosy | Photochemical vs Classical Smog | Stratospheric Ozone Depletion & Free Chlorine Radicals (Montreal Protocol) | Greenhouse Gases & Infrared Absorption (Argon is NOT a GHG) | Heavy Metal Toxicology (Minamata Hg, Itai-Itai Cd, Blackfoot As, Blue Baby Nitrate)

Sources baked in: NCERT Class 9, 11 & 12 (Nuclear Chemistry; Environmental Chemistry; Radioactivity), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on Radioactivity as a nuclear property, Hydrogen bomb based on fusion (not fission), Dynamite invented by Alfred Nobel (nitroglycerin), RDX chemical identity, Argon not being a greenhouse gas, Acid rain caused by SO_2 and NO_2, and Minamata disease by Mercury. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Global Carbon & Methane Pledges**	Enhanced monitoring of non-CO_2 greenhouse gases under COP agreements. Methane (CH_4) possesses a **Global Warming Potential (GWP) \approx 28–36\text{ times} greater than CO_2** over a 100-year timescale, primarily emanating from flooded paddy cultivation, livestock rumination, and coalbed degassing.
2025	**UPPCS Prelims 2025**	Direct question on **Stratospheric Ozone Layer & CFCs**: Both statements confirmed correct—the ozone layer shields the biosphere from lethal solar UV-B rays, and chlorofluorocarbons (CFCs) trigger catalytic ozone destruction via free chlorine radicals (Cl^\bullet).
2024	**UPPCS Prelims 2024**	Assertion-Reason on **Acid Rain Chemistry**: Rainwater acidity is driven by atmospheric SO_2 and NO_x dissolving to form sulphuric (H_2SO_4) and nitric (HNO_3) acids, with pH dropping below 5.6. Causes severe "stone cancer" on white marble monuments (CaCO_3 \to CaSO_4).
2022	**UPPCS Prelims 2022**	Greenhouse Gas Identification: **Argon (Ar)** is strictly **NOT** a greenhouse gas. Diatomic/monatomic symmetric gases (N_2, O_2, Ar) lack dipole moments during vibrational excitation and cannot absorb infrared radiation.

Year	Event	Core Concept & Must-Score Significance
2021	**UPPCS Prelims 2021**	Nuclear vs Atomic Trap: **Radioactivity is strictly a nuclear property** (originating in the unstable nucleus), completely unaffected by chemical bonds, temperature, or pressure. Confirmed the **Hydrogen Bomb operates on Nuclear Fusion**, using a fission device only as a detonating trigger.

📋 **Master Reference Tables (Click to Expand)**

Table 1: Master Classification of High Chemical Explosives | Explosive Name / Acronym | Systematic Chemical Identity & Formula | Key Raw Materials / Synthesis | Detonation Characteristics | Core Applications & Must-Score Signatures | |---|---|---|---|---| | **Dynamite** | **Nitroglycerin (GTN)** absorbed in Kieselguhr / wood pulp | Glycerol treated with concentrated $HNO_3 + H_2SO_4$ | Shock-insensitive, stable | Invented by **Alfred Nobel (1867)**; civil mining and railway tunneling. | | **TNT** | **2,4,6-Trinitrotoluene** ($C_6H_2(CH_3)(NO_2)_3$) | Nitration of toluene with mixed nitric-sulphuric acid | Detonation velocity $\approx 6900\text{ m/s}$ | Military artillery shells; standard reference benchmark for nuclear weapon yields. | | **RDX (Cyclonite / Hexogen / T4)** | **Cyclotrimethylenetrinitramine** ($C_3H_6N_6O_6$) | Nitration of hexamine (urotropine) with fuming HNO_3 | Detonation velocity $\approx 8750\text{ m/s}$ | Highly stable; blended with plasticizers to produce **C-4 Plastic Explosive**. | | **PETN** | **Pentaerythritol Tetranitrate** ($C_5H_8N_4O_{12}$) | Nitration of pentaerythritol | High brisance, shock-sensitive | Used in military detonator cords (Primacord) and explosive boosters. | | **ANFO** | **Ammonium Nitrate Fuel Oil** ($94\%\ NH_4NO_3 + 6\%\text{ oil}$) | Mixing industrial fertilizer prills with diesel | Requires high-explosive booster | Low-cost blasting agent for open-cast coal mining and rock quarrying. | | **Gunpowder (Black Powder)** | **Potassium Nitrate (75%) + Charcoal (15%) + Sulphur (10%)** | Physical mechanical grinding | Low explosive deflagration | Historical firearms propellant; modern fireworks, safety fuses. | --- ### Table 2: Heavy Metal Water & Soil Toxicity Diseases | Disease / Syndrome Name | Causative Toxic Chemical / Metal | Environmental Source / Route of Exposure | Target Organs Affected | Diagnostic Clinical Symptoms & Must-Score | |---|---|---|---|---| | **Minamata Disease** | **Methylmercury (CH_3Hg^+)** | Ingestion of contaminated fish from industrial bay | Central Nervous System, brain cortex | Severe ataxia, numbness in limbs, tunnel vision, deafness, death (Japan, 1956). | | **Itai-Itai Disease** | **Cadmium (Cd)** | River water contaminated by zinc/lead mines irrigating rice | Skeletal bones, kidney renal tubules | **Osteomalacia**, bone softening, multiple painful fractures, kidney failure. | | **Blackfoot Disease** | **Inorganic Arsenic (As)** | Contaminated groundwater in alluvial plains (Bengal basin) | Peripheral vascular system, skin | Gangrenous necrosis of lower extremities, hyperkeratosis of palms, skin cancer. | | **Blue Baby Syndrome** | **Nitrate ($NO_3^- > 45\text{ mg/L}$)** | Runoff of agricultural fertilizers into drinking wells | Blood hemoglobin | Oxidizes $Fe^{2+} \to Fe^{3+}$, forming **methemoglobin**; infant cyanosis (hypoxia). | | **Dental & Skeletal Fluorosis** | **Fluoride ($F^- > 1.5\text{ ppm}$)** | Deep granitic groundwater leaching natural fluoride | Tooth enamel, skeletal bone matrix | Enamel yellow-brown mottling; permanent skeletal stiffness, knock-knee deformity. | | **Plumbism (Saturnism)** | **Lead (Pb)** | Leaded paints, old lead plumbing pipes, industrial fumes | Central nervous system, bone marrow | Lowered IQ in children, anemia, behavioral changes, **Burton's blue line on gums**. |

Detailed Theory Notes

8.1 Radioactivity Kinetics & Radiometric Dating Principles

1. Fundamental Decay Physics

- Unstable nuclei with atomic numbers Z > 82 (beyond Lead) possess excess positive charges causing electrostatic repulsion that overcomes the attractive strong nuclear force.
- The decay rate is purely statistical: \frac{dN}{dt} = -\lambda N Integration yields the exponential law: N(t) = N_0 e^{-\lambda t}.
- **Activity (A)**: The rate of nuclear disintegrations per unit time (A = \lambda N). Measured in Becquerels (\text{Bq}).

2. Radiocarbon Dating (^{14}C)

- Natural nitrogen in the upper atmosphere is bombarded by cosmic-ray neutrons: ^{14}_7N + {}^1_0n \to {}^{14}_6C + {}^1_1p
- ^{14}C rapidly oxidizes into ^{14}CO_2, which enters the terrestrial carbon cycle via plant photosynthesis and food chains.
- While an organism is alive, continuous metabolic exchange maintains a constant equilibrium ratio of ^{14}C / {}^{12}C \approx 1 : 10^{12}.
- Upon death, metabolic assimilation ceases. The radioactive ^{14}C within tissues disintegrates with a half-life of **5,730\text{ years}**: {}^{14}*6C \xrightarrow{\beta^-} {}^{14}_7N + {}^0*e
- Measuring the remaining ^{14}C / {}^{12}C ratio via accelerator mass spectrometry enables precise dating of organic relics up to \approx 50,000\text{ years}.

8.2 Nuclear Weapons & Reactor Core Physics

1. Fission vs. Fusion Energy Dynamics

- **Nuclear Fission**: Heavy nuclei (A \approx 235) have a binding energy per nucleon of \approx 7.6\text{ MeV}. Splitting into fragments (A \approx 120) with higher binding energy (\approx 8.5\text{ MeV}) releases \approx 0.9\text{ MeV} per nucleon, or \approx 200\text{ MeV} per fission event.
- **Nuclear Fusion**: Light nuclei (^2H, ^3H) with low binding energy (\approx 1.1\text{ to }2.8\text{ MeV}) fuse into tightly bound ^4He (\approx 7.07\text{ MeV/nucleon}), releasing \approx 17.6\text{ MeV} for only 5 nucleons, yielding \approx 3.5\text{ MeV} per nucleon. Thus, **gravimetrically, nuclear fusion yields roughly four times more energy per kilogram of fuel than nuclear fission**.

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[ Nuclear Fission ]                          [ Nuclear Fusion ]
Splits heavy nuclei (U-235, Pu-239)          Fuses light nuclei (Deuterium + Tritium)
Triggered by slow thermal neutrons            Triggered by extreme heat (>10⁷ K) & pressure
Used in Nuclear Reactors & Atomic Bombs       Powers the Sun & Hydrogen Bombs
High radioactive fission waste products       Minimal direct radioactive waste products
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8.3 Atmospheric Chemistry: Acid Rain & Ozone Depletion

1. Acid Rain Dynamics

- High-temperature combustion in automobile engines oxidizes atmospheric nitrogen to nitric oxide: $N_2 + O_2 \xrightarrow{>1500^\circ\text{C}} 2NO, \quad 2NO + O_2 \to 2NO_2$
- Coal combustion oxidizes pyrite and organic sulphur impurities: $S + O_2 \to SO_2$
- In the presence of suspended particulate matter (soot, metal dust) and sunlight, SO_2 and NO_2 undergo aqueous-phase catalytic oxidation in cloud droplets into H_2SO_4 and HNO_3.
- Marble stone corrosion (Stone Leprosy): $CaCO_3\text{ (Insoluble marble)} + H_2SO_4 \to CaSO_4 \cdot 2H_2O\text{ (Soluble gypsum)} + CO_2 \uparrow$

2. Stratospheric Ozone Destruction by CFCs

- The stability of CFCs enables them to survive intact in the troposphere for 50–100 years.
- In the stratosphere, solar UV radiation photolyzes CFCs to release atomic chlorine: $\text{Initiation: } CF_2Cl_2 \xrightarrow{h\nu\ (\lambda < 220\text{ nm})} CF_2Cl^\bullet + Cl^\bullet \text{Propagation 1: } Cl^\bullet + O_3 \to ClO^\bullet + O_2 \text{Propagation 2: } ClO^\bullet + O^\bullet \to Cl^\bullet + O_2 \text{Net Reaction: } O_3 + O^\bullet \to 2O_2$
- The $Cl^\bullet$ radical acts as a true homogeneous catalyst: it accelerates the destruction of ozone without being consumed in the net reaction.

Bilingual Terminology -- Radiation, Nuclear and Environmental Chemistry

English Term	Hindi Term	Key Anchor
Radiation (विकिरण)	विकिरण	Energy emitted as electromagnetic waves or particles
Ionising Radiation (आयनकारी विकिरण)	आयनकारी विकिरण	High-energy radiation that removes electrons from atoms; alpha, beta, gamma, X-ray
Non-ionising Radiation (अआयनकारी विकिरण)	अनायनकारी विकिरण	Lower energy; radio waves, microwaves, visible light, UV-A
Radioactive Waste (रेडियोधर्मी अपशिष्ट)	रेडियोधर्मी अपशिष्ट	Waste from nuclear reactors, hospitals; requires careful disposal
Half-life (अर्ध-जीवन)	अर्ध-जीवन	Time for half of radioactive atoms to decay
Radiocarbon Dating (रेडियोकार्बन काल-निर्धारण)	रेडियोकार्बन डेटिंग	Uses C-14 decay to date organic material up to ~50,000 years
UV Radiation (पराबैंगनी विकिरण)	पराबैंगनी विकिरण	Electromagnetic radiation 10-400 nm; UV-B causes sunburn; blocked by ozone
Infrared Radiation (अवरक्त विकिरण)	अवरक्त / ताप विकिरण	Electromagnetic radiation 700 nm-1 mm; heat radiation
Microwave (सूक्ष्म तरंग)	सूक्ष्म तरंग	1 mm-1 m wavelength; RADAR; microwave oven (2.45 GHz)

English Term	Hindi Term	Key Anchor
Air Pollution (वायु प्रदूषण)	वायु प्रदूषण	Contamination of air with harmful gases/particles
Water Pollution (जल प्रदूषण)	जल प्रदूषण	Contamination of water bodies
Acid Rain (अम्ल वर्षा)	अम्लीय वर्षा	pH < 5.6; H2SO4 + HNO3 from SO2 + NOx emissions
Eutrophication (सुपोषण)	पोषकता वृद्धि	Excess nutrients in water -> algae bloom -> O2 depletion
BOD (जैव रासायनिक ऑक्सीजन मांग)	जैव रासायनिक ऑक्सीजन माँग	Biochemical Oxygen Demand; measures organic water pollution
Photochemical Smog (प्रकाश रासायनिक धुंध)	प्रकाश-रासायनिक स्मॉग	NO + VOC + sunlight -> ozone + PAN; brown smog; Los Angeles type
e-Waste (इलेक्ट्रॉनिक कचरा)	इलेक्ट्रॉनिक कचरा	Discarded electronic devices; contains toxic metals (Pb, Cd, Hg)

One-Liner Revision -- Radiation, Nuclear and Environmental Chemistry

#	Fast Fact
1	Ionising radiation: alpha, beta, gamma, X-rays, cosmic rays -- can cause cancer
2	Non-ionising: radio, microwave, visible, UV-A -- generally safer (UV-B intermediate)
3	UV-B (280-315 nm) = causes sunburn, skin cancer; blocked by ozone layer
4	UV-A (315-400 nm) = causes tanning; glass blocks UV; used in sterilisation
5	Microwave oven = 2.45 GHz; water molecules absorb and vibrate -> heat food
6	C-14 half-life = **5730 years**; used for dating organic material
7	Radioactive decay: alpha (weakest penetration), beta, gamma (strongest penetration)
8	Radiation doses: mSv (millisievert); background dose ~2-3 mSv/year
9	Radiation therapy = high-dose gamma/X-ray to kill cancer cells
10	Chernobyl (1986) = RBMK reactor explosion; released Cs-137 and I-131
11	Acid rain: SO2 + H2O -> H2SO3; SO3 + H2O -> H2SO4; causes damage to Taj Mahal marble
12	Acid rain pH < 5.6 (normal rain = 5.6 due to natural CO2)
13	London smog = industrial smog (SO2 + soot + fog); cold; 1952 Great Smog killed 4,000+
14	LA smog = photochemical smog (NOx + VOC + sunlight -> O3 + PAN); warm, sunny
15	Ozone (O3) in troposphere = pollutant (causes respiratory problems); in stratosphere = protective
16	PM2.5 = particulate matter < 2.5 micron; deepest lung penetration; most harmful
17	PM10 = particles < 10 micron; cause respiratory problems

#	Fast Fact
18	BOD: high BOD = more organic pollution; clean water BOD < 3 mg/L
19	COD (Chemical Oxygen Demand) = total oxygen needed to chemically oxidise all organics
20	Eutrophication: excess N and P (from fertiliser runoff) -> algal bloom -> dead zone
21	Biomagnification = persistent pollutants (DDT, Hg, PCBs) concentrate up the food chain
22	Mercury (Hg) poisoning = Minamata disease (Japan, 1950s); methylmercury in fish
23	Cadmium poisoning = Itai-itai disease (Japan); bone pain; from mine runoff
24	Lead (Pb) poisoning = cognitive damage in children; banned from petrol (TEL) and paint
25	Arsenic contamination = groundwater in Bangladesh, West Bengal; skin lesions, cancer
26	Fluorosis = excess fluoride (>1.5 ppm) = mottled teeth (dental); skeletal damage (skeletal)
27	e-waste = toxic heavy metals; India = 3rd largest e-waste generator globally
28	Basel Convention (1989) = controls transboundary movement of hazardous wastes
29	Stockholm Convention (2001) = bans persistent organic pollutants (POPs) like DDT, PCBs
30	CPCB = Central Pollution Control Board (India); SPCB = State Pollution Control Board